Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/17/2005
PI Name:	LiCata, Vince J Ph.D.		
Project Title:	The Effects of Microgravity on Protein-Ligand Interactions		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Molecular structures and physical interactions		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	licata@lsu.edu	**Fax:**	FY 225-578-2597
PI Organization Type:	UNIVERSITY	**Phone:**	225-578-5233
Organization Name:	Louisiana State University		
PI Address 1:	Department of Biological Sciences		
PI Address 2:	202 Life Sciences Building		
PI Web Page:	http://www.biology.lsu.edu/		
City:	Baton Rouge	**State:**	LA
Zip Code:	70803	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	08/01/2001	**End Date:**	07/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 2-1511		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In this proposal, several specific chemical kinetics and protein-ligand systems will be examined in parallel in ground based versus reduced gravity experiments: 1) the association of adipocyte lipid binding protein (ALBP) with lipids; 2) the association of bovine serum albumin and lipids; and 3) the association of DNA polymerase 1 (Pol 1) with DNA; and the reaction of ascorbic acid and DCIP as a function of pH. Parallel kinetic and equilibrium binding assays will be carried out in all three systems in the laboratory and in reduced gravity during parabolic flights on NASA's KC-135A aircraft. Assays will also be performed as a function of viscosity to quantitate the gravitational effect on the degree of diffusion control of each interaction.. Because of the established relationship between electrostatic attraction and the rates of diffusion of charged substrates toward enzymes, the effects of altering the electrostatics of the protein-ligand attractions in these systems will also be examined.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal examines the effects of microgravity on protein-ligand interactions, with a focus on fast and diffusion controlled reactions. Each of the biochemical reactions to be studied is physiologically important: 1) adipocyte lipid binding protein (ALBP) is linked to both obesity and the development of Type II diabetes; 2) serum albumin is responsible for transport of fatty acids and other lipids in the bloodstream; 3) Pol 1 is involved in DNA replication and repair in vivo, and is a model system for understanding DNA polymerases in general. Microgravity effects on any of these reactions would have important physiological consequences. The main focus of this proposal, however, is to utilize these systems as benchmark reactions to test the effects of microgravity on protein-ligand interactions in general. Microgravity, with it's reduced convective mixing and lack of molecular buoyancy or sedimentation, is clearly likely to alter the rates, and possibly the equilibria of protein-ligand interactions, especially of fast and diffusion limited reactions. Even relatively small changes in these equilibria or rates could have significant physiological effects. Gravitational effects may be wholly or partially responsible for the fact that even the fastest known enzymes never attain the theoretical limit for diffusion controlled reactions.
Task Progress:	One set of flights was completed on NASA's KC-135. On this flight set kinetic reactions of ascorbic acid and DCIP (2,6 dichloroindophenol) were characterized at several pH's. Preliminary characterization of binding of ANS (1,8 anilinonaphthalene sulfonate) to BSA (bovine serum albumin) was also performed, but experiments could not be completed due to cancellation of the last day of flights. Changeover of airplanes in the JSC Reduced Gravity Program has resulted in a (currently) 7 month hiatus in reduced gravity flights. As a result this grant was no cost extended into 2005, and currently microgravity flights are scheduled for later in 2005 to complete this portion of the specific aims.
Bibliography Type:	Description: (Last Updated: 01/31/2008)
Abstracts for Journals and Proceedings	Schoeffler, A.J., Joubert, A.M., Solhjoo, L., Yang, X., and LiCata, V.J. "Salt effects on the lipid binding and stability of adipocyte lipid binding protein" Meeting abstract Biophysical Journal, Vol 84, p. 333A , Feb-2003
Articles in Peer-reviewed Journals	Schoeffler, A.J., Ruiz, C.R., Joubert, A.M., Yang, X., and LiCata, V.J. "Salt modulates the stability and lipid binding affinity of the adipocyte lipid binding protein" Journal of Biological Chemistry Vol. 278, pp. 33268-33275 , Aug-2003